Solution:
The given strings contains 6 letters (V,O,W,E,L,S).

(i) Since all strings must begin with E, we have the remaining 5 letters which can be arranged in ${}^5P_5 = 5!$ ways
Therefore the total number of strings with E as the starting letter is $5! = 120$.

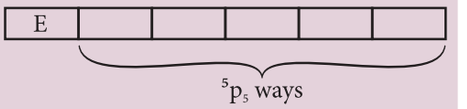


Figure 4.10

(ii) Since all strings must begin with E, and end with W, we need to fix E and W. The remaining 4 letters can be arranged in ${}^4P_4 = 4!$ ways.

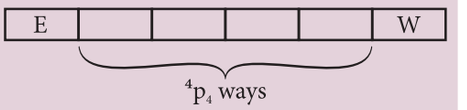


Figure 4.11

Therefore the total number of strings with E as the starting letter and W as the final letter is $4! = 24$.

Example 4.29 A four digit number with four different digits using 1,2,3,4 and 5 is formed Find the following

(i) How many such numbers can be formed?
(ii) How many of these are even?
(iii) How many of these are exactly divisible by 4?

Solution:

(i) The solution for this is the same as the number of permutations taking four digits out of 5 digits is ${}^5P_4 = 5\times4\times3\times2 = 120$.

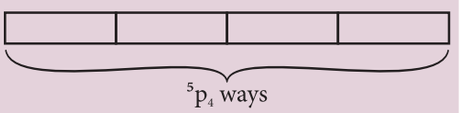


Figure 4.12

(ii) For even number last digits must be 2 or 4 which is filled in ${}^2P_1$ ways and remaining 3 places filled from remaining 4 digits in ${}^4P_3$ ways. Therefore the required number of ways is ${}^2P_1 \times {}^4P_3 = 2 \times 24 = 48$.

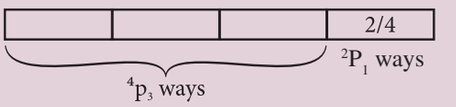


Figure 4.13

(iii) Since the number divisible by 4, then last two digit must be divisible by 4. The Last two digits become 12,24,32,52 (4 ways). The remaining first two places filled from remaining 3 digits in ${}^3P_2$ ways. The required number of numbers which are divisible by 4 is ${}^4P_1 \times {}^3P_2 = 4 \times 6 = 24$.

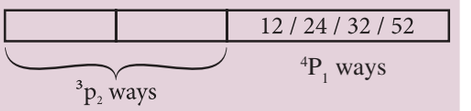


Figure 4.14

4.4.3 Objects always together (String method)

The number of permutations of n different objects, taken all at a time, when m specified objects are always together,